Advait Paranjpe

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Education

University of California, Los Angeles | *M.S. in Electrical and Computer Engineering* Sep 2025 – Dec 2026

- Coursework: CPU/GPU Architecture, Memory Systems, Cache Coherence, Interconnect Design.
- GPA: 3.7/4.0

The University of Manchester | *B.Eng. in Electrical and Electronic Engineering* Sep 2022 – Jul 2025

- Dissertation: "Cascaded PLL Microwave Synthesizer for 5G systems."
- **First Class Honours** – highest UK degree classification.

Experience

Jaguar Land Rover North America | *Incoming Semiconductor Applications Engineering Intern* Jul 2026 – Sep 2026

VAST Lab, UCLA | *Student Researcher – GPU/FPGA Acceleration* Jan 2026 – Mar 2026

- Profiled ROCm attention kernels and memory bottlenecks to identify FPGA acceleration opportunities for Engram LLM inference.
- Modeled FPGA cache architectures, increasing projected hit rate from ∼5% at 256 entries to ∼48% with a 64k-entry, 4-way design.
- Translated workload profiling into FPGA tradeoffs across cache capacity, lookup throughput, locality, and on-chip memory usage.

NESO / National Grid ESO | *Software Engineering Intern (two summers)* Jul 2024 – Sep 2025

- Re-architected a 100k+ LOC simulation engine through vectorization and pipeline optimization, cutting runtime by 10x.
- Optimized 100M+ records through chunked, memory-efficient execution, cutting RAM from 32 GB to 16 GB and runtime by 11x.
- Reduced CPU context-switching overhead through parallel API batching and staged processing, increasing throughput by 24x for time-sensitive market telemetry.

Projects

JobAgent | *Full-Stack Job Application Platform (FastAPI, React, SQLAlchemy)* May 2026 – Jun 2026

- Piloted JobAgent with 10 beta users across 200 scored jobs and 500 generated CV packets, using feedback to refine job capture, scoring, and application workflows.
- Built a FastAPI/SQLAlchemy platform with async packet generation, duplicate-run protection, progress polling, signed-token authentication, user isolation, quotas, and artifact validation.
- Shipped a Chrome MV3 extension and React tracker for job capture, scoring, PDF preview/download, and status updates, backed by 192 automated pytest cases covering core workflows and security controls.

Agentic RTL Security Discovery | *Simulation and Test Automation Tooling (Python, SystemVerilog)* May 2026

- Built a Python/Icarus Verilog harness that executes generated MMIO traces against clean and bug-enabled RTL with oracle-gated validation.
- Implemented an agentic loop for planning, generating, simulating, observing, refining, minimizing, and reporting verification tests.
- Compared deterministic regression, random fuzzing, and LLM-assisted discovery across bug detection, reproducibility, and manual effort.

SparrowML | *ML Compiler and Runtime Toolchain (Python, C, PyTorch)* May 2026 – Jun 2026

- Built a modular pipeline for model training, INT8 quantization, tensor serialization, C runtime generation, and deterministic target packaging.
- Generated code, packed tensor layouts, and workloads for scalar, dense-vector, sparse-vector, and multicore RISC-V execution.
- Added reproducible evaluation and benchmarking across prediction quality, cycles, memory traffic, vector utilization, and multicore scaling.

Skills

- **Languages:** Python, C++, C, SQL, Bash, JavaScript, TypeScript, MATLAB, SystemVerilog.
- **Backend and Full Stack:** FastAPI, SQLAlchemy, REST APIs, React, Chrome MV3, asynchronous workflows, data pipelines, and PDF generation.
- **Engineering Tools:** pytest, Git, Linux, Docker, CI/CD, profiling, multiprocessing, NumPy, Pandas, and HIP/ROCm.